

HIIT Strategies for **20** Sports

Science and
Application of
**HIGH-INTENSITY
INTERVAL
TRAINING**

Solutions to the Programming Puzzle



Paul Laursen
Martin Buchheit
EDITORS

Science and Application of High-Intensity Interval Training



Editors

Paul Laursen, PhD

Martin Buchheit, PhD



HUMAN KINETICS

Library of Congress Cataloging-in-Publication Data

Names: Laursen, Paul, 1971- editor. | Buchheit, Martin, 1978- editor.

Title: Science and application of high-intensity interval training : solutions to the programming puzzle / Paul Laursen, Martin Buchheit, editors.

Description: Champaign, IL : Human Kinetics, [2018] | Includes bibliographical references and index.

Identifiers: LCCN 2018030511 (print) | LCCN 2018044768 (ebook) | ISBN 9781492586890 (ebook) | ISBN 9781492552123 (print)

Subjects: LCSH: Interval training. | Athletes—Training of.

Classification: LCC GV481 (ebook) | LCC GV481 .S333 2018 (print) | DDC 613.7/04—dc23

LC record available at <https://lcn.loc.gov/2018030511>

ISBN: 978-1-4925-5212-3 (print)

Copyright © 2019 by Paul Laursen and Martin Buchheit

All rights reserved. Except for use in a review, the reproduction or utilization of this work in any form or by any electronic, mechanical, or other means, now known or hereafter invented, including xerography, photocopying, and recording, and in any information storage and retrieval system, is forbidden without the written permission of the publisher.

The web addresses cited in this text were current as of October 2018, unless otherwise noted.

Senior Acquisitions Editor: Roger W. Earle

Senior Developmental Editor: Cynthia McEntire

Indexer: Rebecca McCorkle

Permissions Manager: Martha Gullo

Graphic Designer: Whitney Milburn

Cover Designer: Keri Evans

Cover Design Associate: Susan Rothermel Allen

Photo Asset Manager: Laura Fitch

Photo Production Manager: Jason Allen

Senior Art Manager: Kelly Hendren

Art Style Development: Joanne Brummett

Illustrations: © Human Kinetics

Production: Westchester Publishing Services

Printer: Walsworth

Human Kinetics books are available at special discounts for bulk purchase. Special editions or book excerpts can also be created to specification. For details, contact the Special Sales Manager at Human Kinetics.

Printed in the United States of America

10 9 8 7 6 5 4 3 2 1

The paper in this book was manufactured using responsible forestry methods.

Human Kinetics

P.O. Box 5076

Champaign, IL 61825-5076

Website: www.HumanKinetics.com

In the United States, email info@hkusa.com or call 800-747-4457.

In Canada, email info@hkcanada.com.

In the United Kingdom/Europe, email hk@hkeurope.com.

For information about Human Kinetics' coverage in other areas of the world, please visit our website: **www.HumanKinetics.com**

E7078

Contents

Part I	Understanding High-Intensity Interval Training	1
1	Genesis and Evolution of High-Intensity Interval Training	3
	<i>Paul Laursen and Martin Buchheit with contributions from Jean Claude Vollmer</i>	
2	Traditional Methods of HIIT Programming	17
	<i>Martin Buchheit and Paul Laursen</i>	
3	Physiological Targets of HIIT	33
	<i>Martin Buchheit and Paul Laursen</i>	
4	Manipulating HIIT Variables	51
	<i>Martin Buchheit and Paul Laursen</i>	
5	Using HIIT Weapons	73
	<i>Martin Buchheit and Paul Laursen</i>	
6	Incorporating HIIT Into a Concurrent Training Program	119
	<i>Jackson Fyfe, Martin Buchheit, and Paul Laursen</i>	

7	HIIT and Its Influence on Stress, Fatigue, and Athlete Health	137
	<i>Philip Maffetone, Paul Laursen, and Martin Buchheit</i>	
8	Quantifying Training Load	161
	<i>Martin Buchheit and Paul Laursen</i>	
9	Response to Load	179
	<i>Martin Buchheit, Paul Laursen, Jamie Stanley, Daniel Plews, Hani Al Haddad, Mathieu Lacome, Ben Simpson, and Anna Saw</i>	
10	Putting It All Together	213
	<i>Paul Laursen and Martin Buchheit</i>	
Part II	Sport-Specific Application of High-Intensity Interval Training	225
11	Combat Sports	227
	<i>Duncan French</i>	
12	Cross-Country Skiing	247
	<i>Øyvind Sandbakk</i>	
13	Middle-Distance Running	261
	<i>Jean Claude Vollmer and Martin Buchheit</i>	
14	Road Running	283
	<i>Jamie Stanley and Carlos Alberto Cavalheiro</i>	

15	Road Cycling	297
	<i>Marc Quod</i>	
16	Rowing	311
	<i>Daniel Plews</i>	
17	Swimming	325
	<i>Tom Vandenbogaerde, Wim Derave, and Philippe Hellard</i>	
18	Tennis	347
	<i>Jaime Fernandez-Fernandez</i>	
19	Triathlon	363
	<i>Daniel Plews and Paul Laursen</i>	
20	American Football	379
	<i>Johann Bilsborough and Moses Cabrera</i>	
21	Australian Football	393
	<i>Aaron Coutts, Joel Hocking, and Johann Bilsborough</i>	
22	Baseball	411
	<i>Robert Butler and Matt Leonard</i>	
23	Basketball	425
	<i>Xavi Schelling and Lorena Torres-Ronda</i>	
24	Cricket	443
	<i>Carl Petersen and Aaron Kellett</i>	

25	Field Hockey	455
	<i>Dave Hamilton</i>	
26	Ice Hockey	477
	<i>Matt Nichol</i>	
27	Handball	495
	<i>Martin Buchheit</i>	
28	Rugby Union	511
	<i>Nic Gill and Martyn Beaven</i>	
29	Rugby Sevens	525
	<i>Nick Poulos</i>	
30	Soccer	547
	<i>Martin Buchheit, Mathieu Lacome, and Ben Simpson</i>	
	References	565
	Index	647
	About the Editors	661
	List of Contributors	663
	Earn Continuing Education Credits/Units	665